



VSTS CONTINUOUS BUILD INTEGRATION FOR MAVEN PROJECTS VERSION 1.0

Date:
March 26, 2020

Presented by:
CrimsonLogic India Pvt Ltd.

DELIVERY HISTORY

Version	Brief Summary o Description of changes	Author	Date
1.0	Initial Delivery	CrimsonLogic	March 26, 2020

PREPARED BY

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REVIEWED BY

ROLE	NAME	SIGNATURE	DATE (DD.MM.YYYY)
Senior Software Engineer (VSTS Admin)	Swagath Chalukya		

RELEASE HISTORY

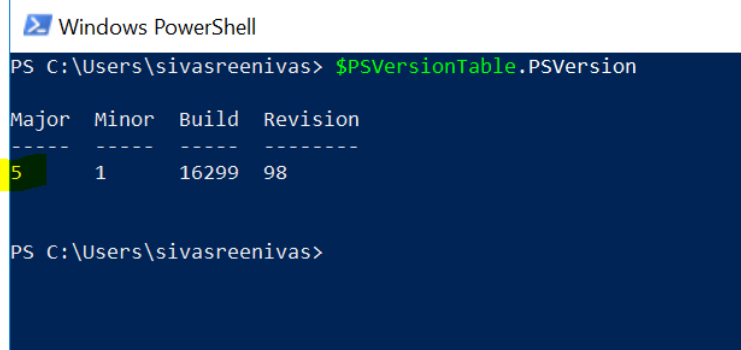
RELEASE NUMBER	ASSOCIATED CHANGE REQUEST NUMBER	DATE (MM,YYYY)	BRIEF SUMMARY OF CHANGES
1.0	NA	26-MAR-2020	Initial Version

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1. PREREQUISITES

1. All the project specific artefacts should be available in the remote system placed for running the build. (Ex: JDK, Maven & other).
2. Need to check the windows remote system has the capabilities and powershell version for the holding the VSTS build.
 - Check the below in the power shell command prompt
 - Search for power shell by pressing the windows button
 - Type the command `$PSVersionTable.PSVersion` --- **The major version should be 3 or higher to hold the VSTS build**



```
Windows PowerShell
PS C:\Users\sivasreenivas> $PSVersionTable.PSVersion

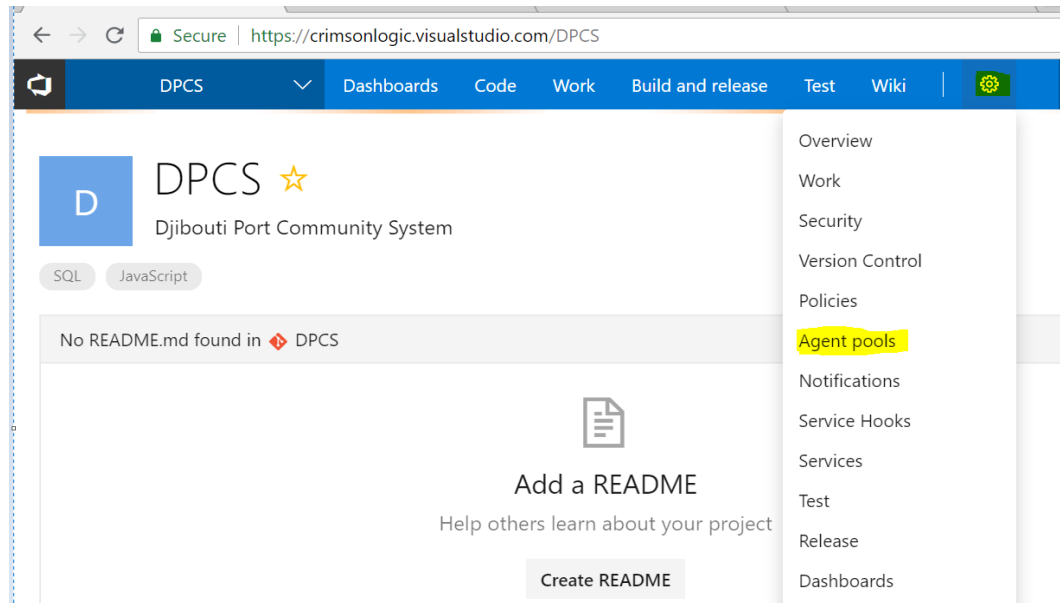
Major Minor Build Revision
-----
5      1      16299 98

PS C:\Users\sivasreenivas>
```

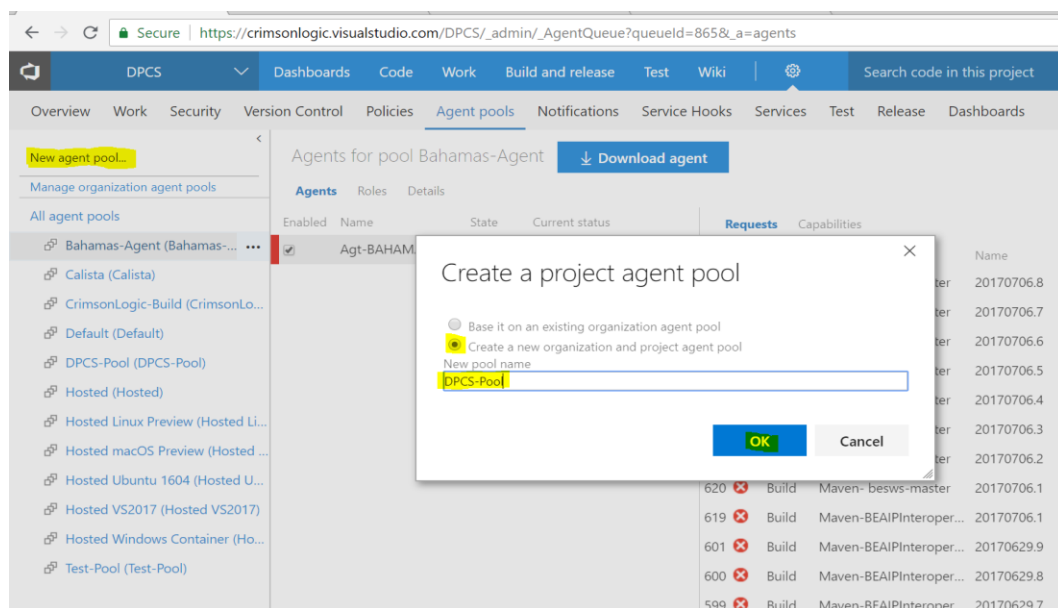
- For more refer the link <https://github.com/Microsoft/azure-pipelines-agent/blob/master/docs/start/envwin.md>
 - In case if the major version is less than 3 please follow the below link to upgrade it to higher version
<https://blogs.technet.microsoft.com/heyscriptingguy/2014/11/09/weekend-scripter-install-powershell-4-0-in-windows-7/>
3. Need to have the build admin permission in the VSTS to trigger the build from VSTS.
 4. The user who is trying to create VSTS agent in the remote system should enable the alternative credentials in the VSTS.

2. STEPS TO CREATE AGENT POOL IN VSTS

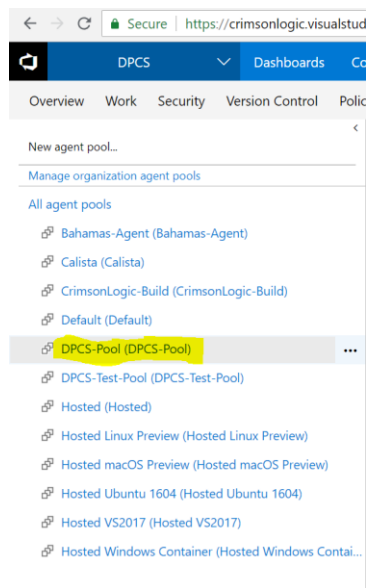
1. Click on gear icon and choose Agent pools from the options listed.



2. Click on **New agent pool..** option available on the top left and *Create a project agent pool pop up* will be opened where need to choose radio button **create a new organization and project agent pool** and enter the name for **New pool name**

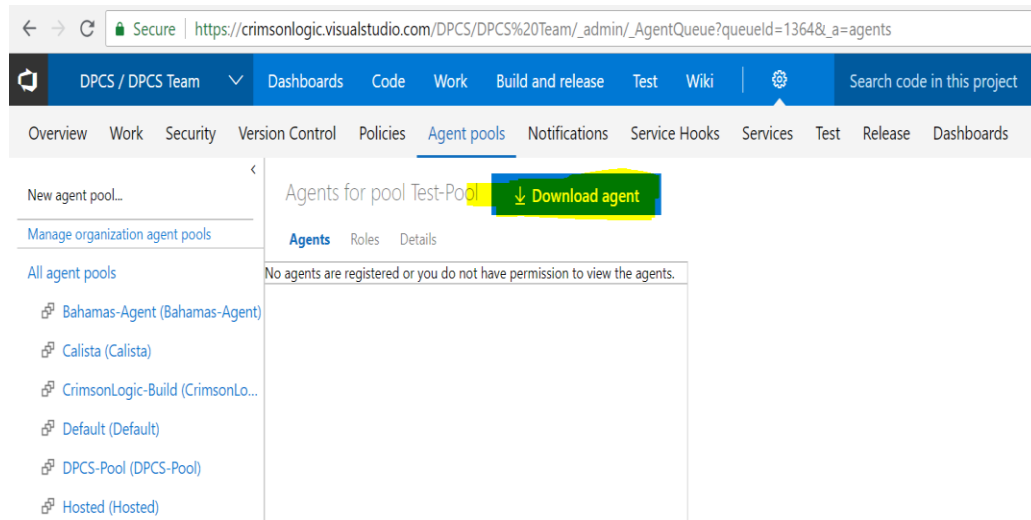


3. After creating the agent pool it will be listed in the left side of the window as shown in below screenshot.

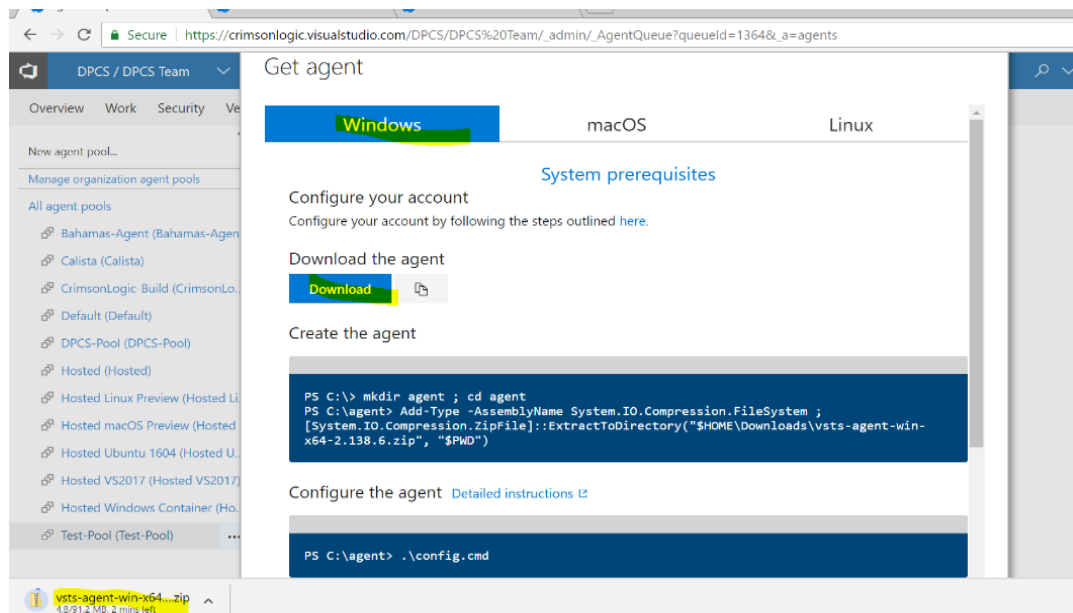


3. STEPS TO CREATE AGENT IN THE REMOTE SYSTEMS

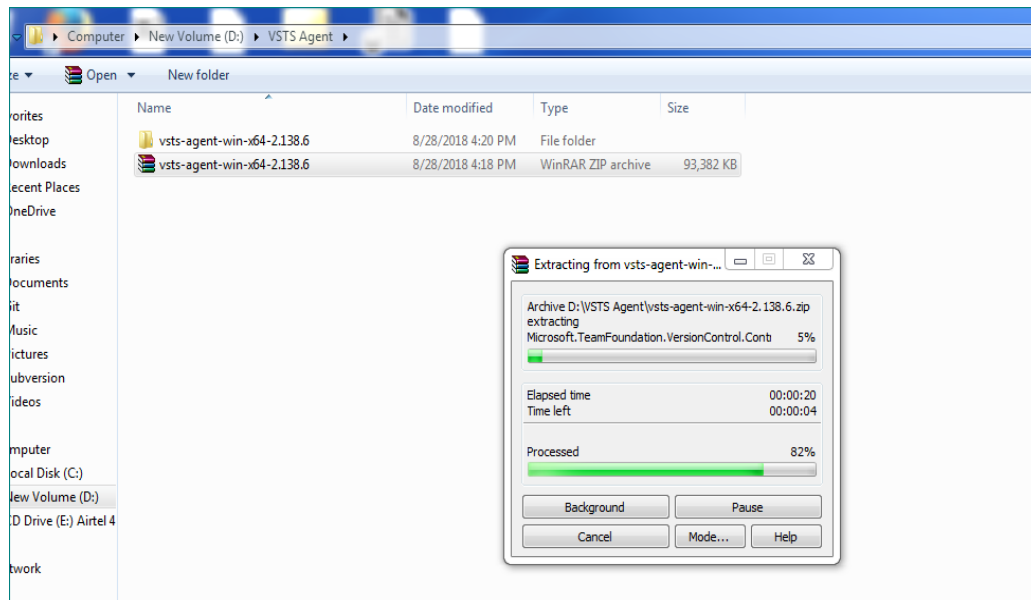
1. Need agent to be downloaded to remote system. Login to VSTS choose the project need to be created the agent. Click on **gear icon** and choose the **Agent pools** option and click on **Download agent** button.



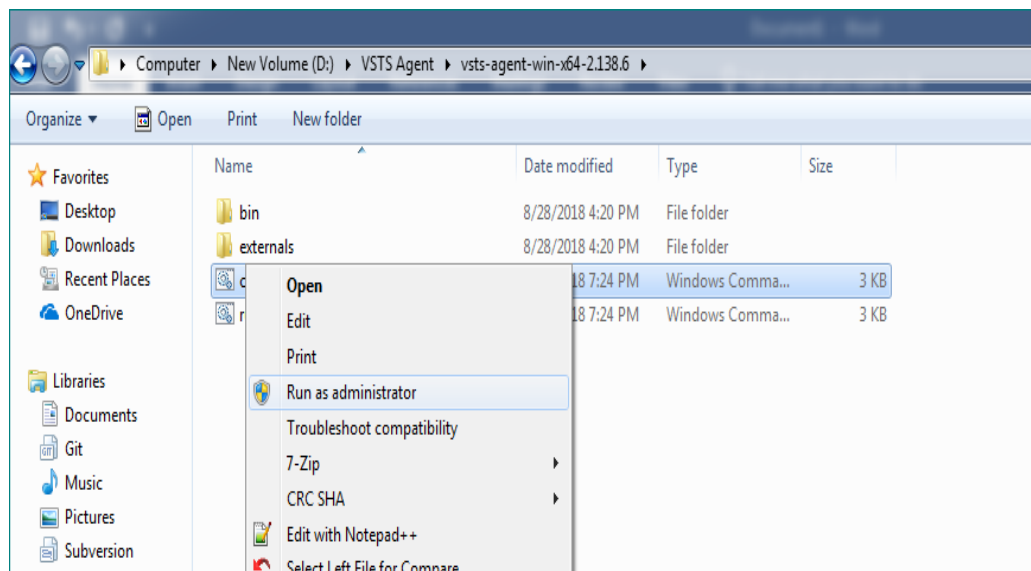
2. A new popup will be opened, choose the download option based on your remote system OS.



3. Create a new folder in any drive, copy and extract the zip folder.

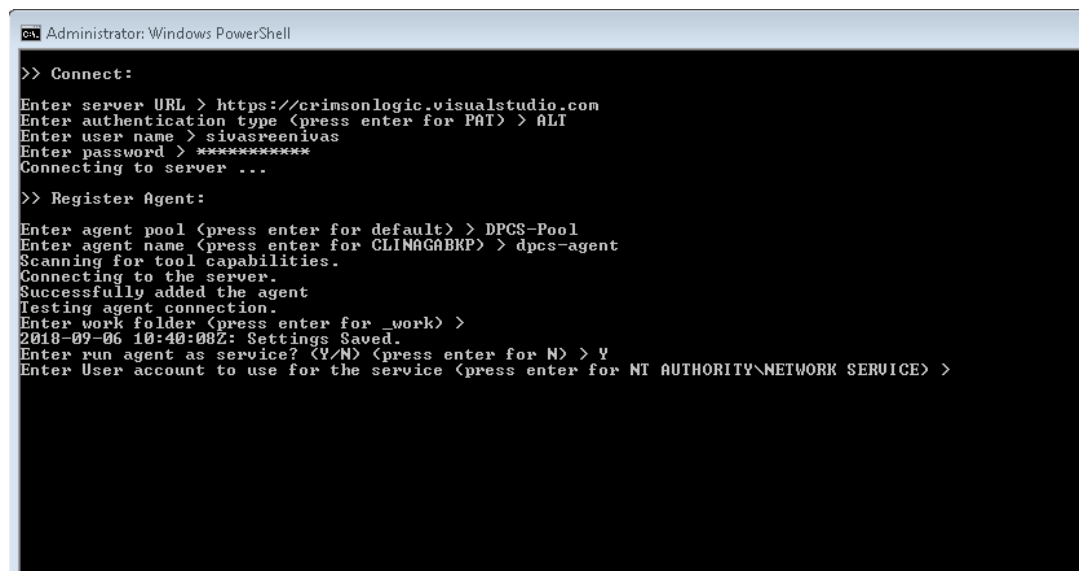


4. Get into the extract folder and run the config batch file in administrator mode.



5. In the command prompt follow the below steps (refer the below screenshot for more details)

```
Enter server URL > https://crimsonlogic.visualstudio.com
Enter authentication type (press enter for PAT) > ALT (opting for alternative credentials)
Enter user name > username (this should be the secondary credentials got for VSTS)
Enter password > *****
Enter agent pool (press enter for default) > DPCS-Pool (pool name should be same what
created above – pool name is case sensitive)
Enter agent name (press enter for *****) > dpcs-agent (can refer to any agent name as
per your project)
Enter work folder (press enter for _work) > skip this to answer by pressing enter
Enter run agent as server? (Y/N) (press enter for N) > Y
Enter User account to use for the service (press enter for NT AUTHORITY\NETWORK SERVICE)
> skip this to answer by pressing enter
```

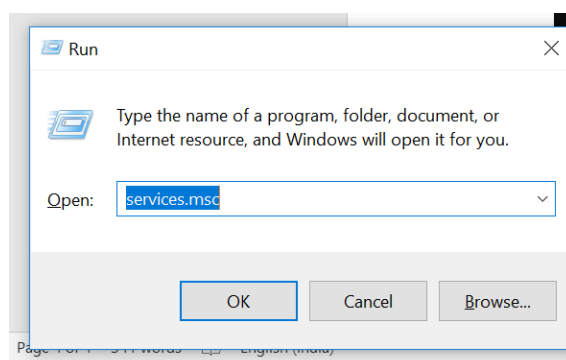


```
Administrator: Windows PowerShell

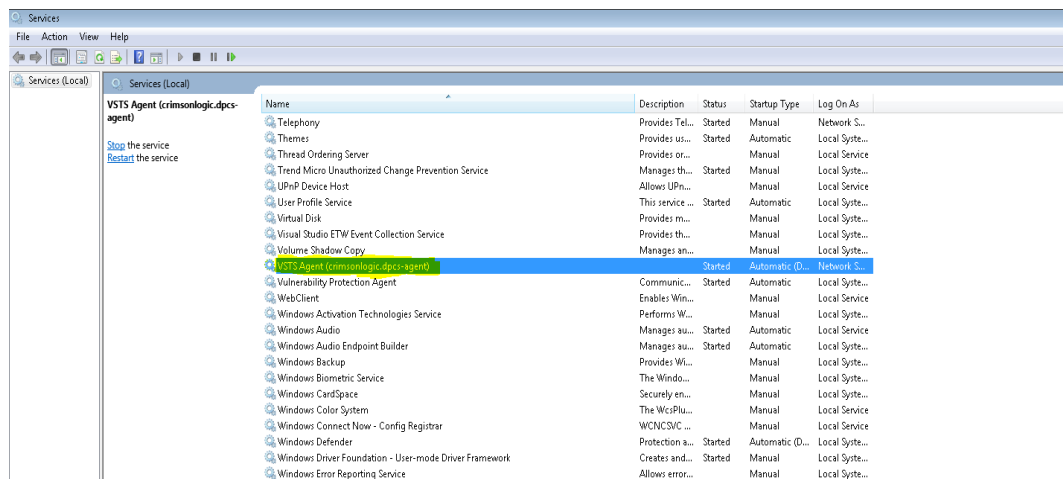
>> Connect:
Enter server URL > https://crimsonlogic.visualstudio.com
Enter authentication type <press enter for PAT> > ALT
Enter user name > sivasreenivas
Enter password > *****
Connecting to server ...

>> Register Agent:
Enter agent pool <press enter for default> > DPCS-Pool
Enter agent name <press enter for CLINAGABKP> > dpcs-agent
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder <press enter for _work> >
2018-09-06 10:40:08Z: Settings Saved.
Enter run agent as service? <Y/N> <press enter for N> > Y
Enter User account to use for the service <press enter for NT AUTHORITY\NETWORK SERVICE> >
```

6. After completing the above steps, the command prompt window will be closed automatically.
7. Once window is closed go to **windows button** → **run** and type **services.msc**

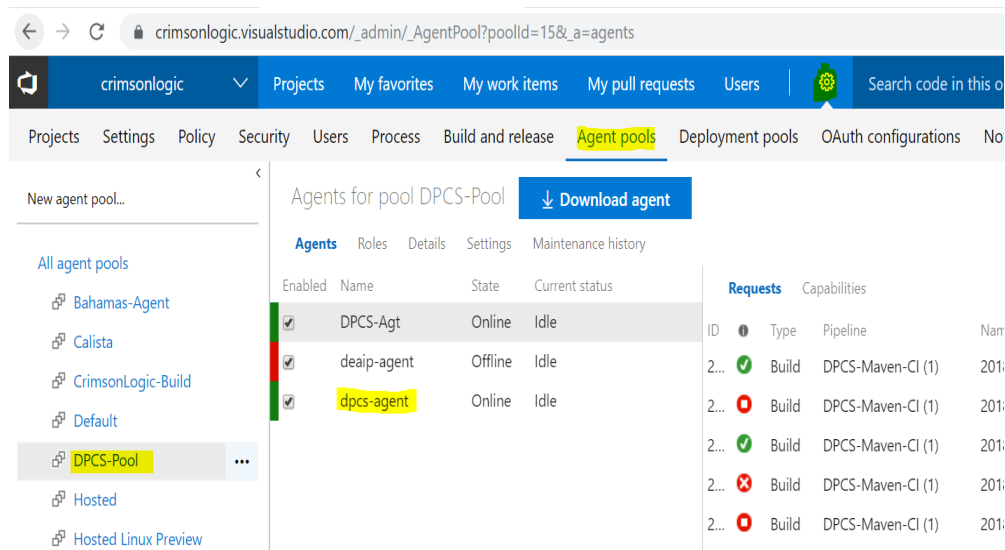


8. Search for VSTS service running and will be found with agent name created in step 5 (if it's not running please start the service)



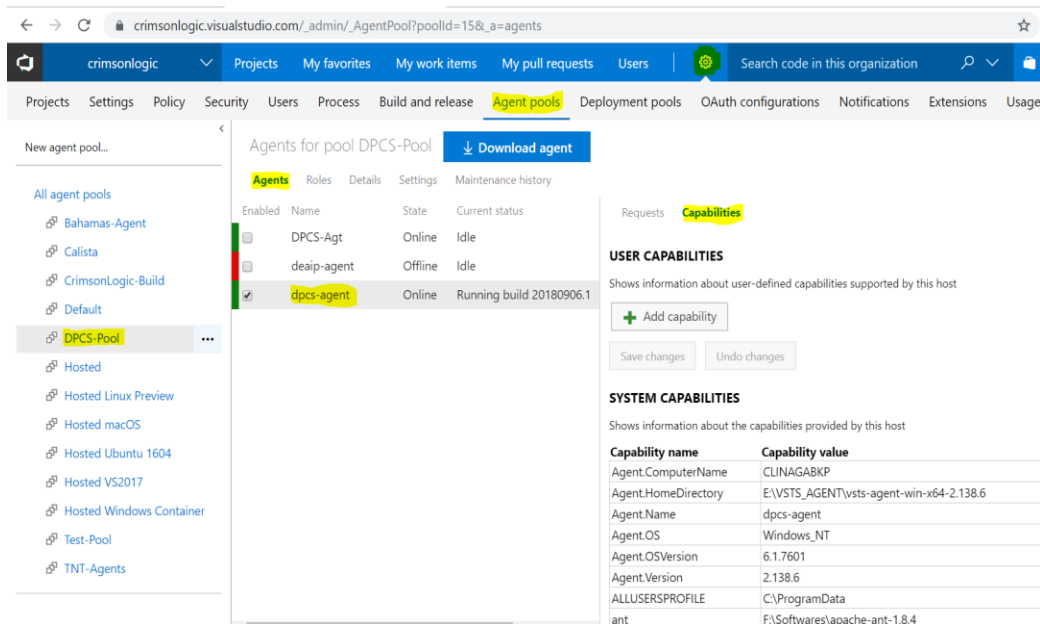
9. After agent service created in remote system is up and running we can check the status of the agent in the VSTS whether its active or inactive.

Gear icon → Agent pools → DPCS-Pool (DPCS-Pool)



10. To verify whether VSTS read all the necessary environment variables of the remote system please check Capabilities tabs by choosing relevant agents.

Gear icon → Agent pools → DPCS-Pool (DPCS-Pool) → Agents → Choose the name (DPCS-Agt) → Capabilities



Agents for pool DPCS-Pool

Enabled	Name	State	Current status
☐	DPCS-Agt	Online	Idle
☐	deaip-agent	Offline	Idle
☒	dpcs-agent	Online	Running build 20180906.1

USER CAPABILITIES

Shows information about user-defined capabilities supported by this host

+ Add capability

Save changes Undo changes

SYSTEM CAPABILITIES

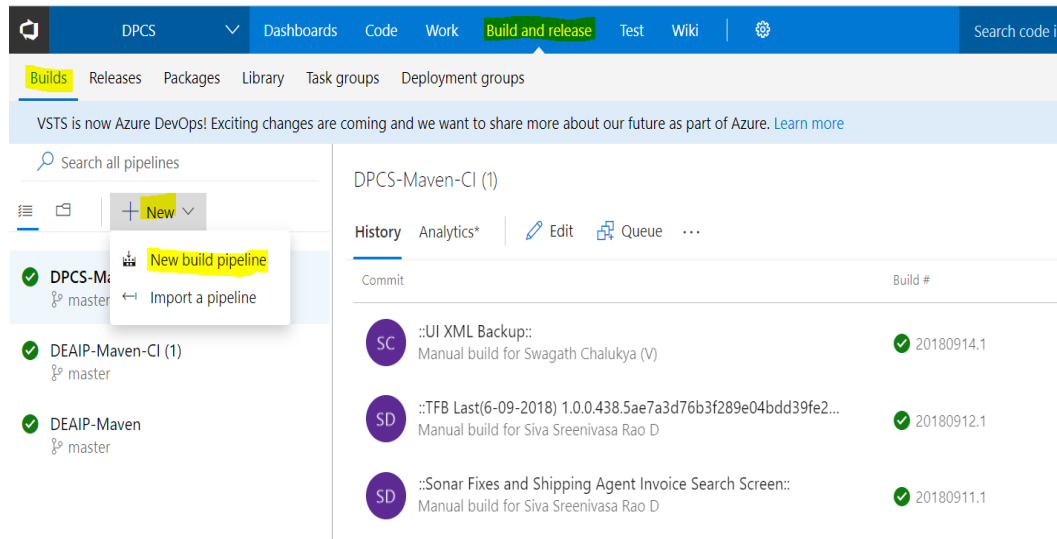
Shows information about the capabilities provided by this host

Capability name	Capability value
Agent.ComputerName	CLINAGABKP
Agent.HomeDirectory	E:\VSTS_AGENT\vsts-agent-win-x64-2.138.6
Agent.Name	dpcs-agent
Agent.OS	Windows_NT
Agent.OSVersion	6.1.7601
Agent.Version	2.138.6
ALLUSERSPROFILE	C:\ProgramData
ant	F:\Softwares\apache-ant-1.8.4

NOTE: Multiple agents can be created under one agent pool. Can enable only the agent which needs to be used for building the project repository. (Refer in the above screenshot)

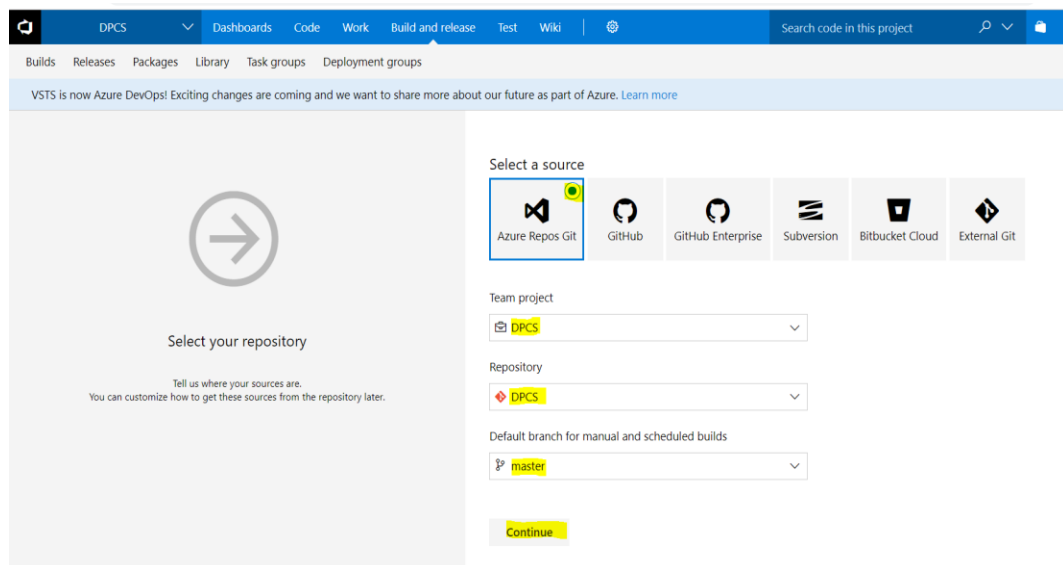
4. STEPS TO CREATE JOBS IN VSTS

1. Click on **Build and release** → **Builds** → **New** → **New build pipeline**



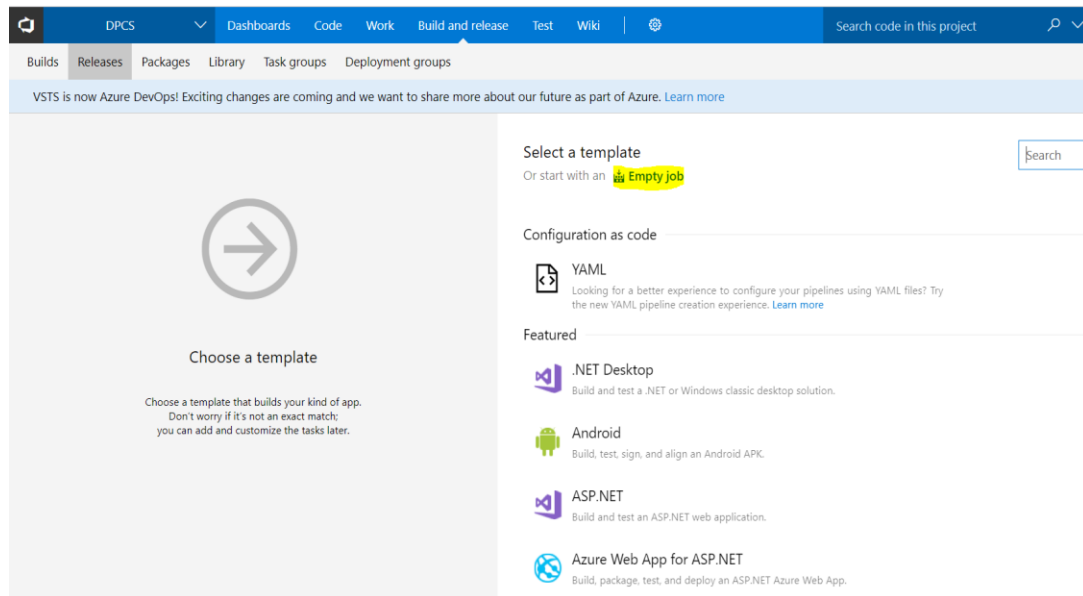
The screenshot shows the VSTS interface with the 'Build and release' tab selected. The 'Builds' sub-tab is active, and the 'New' button is highlighted. A dropdown menu is open, showing 'New build pipeline' as the selected option. The main content area displays a list of build pipelines for 'DPCS-Maven-CI (1)', including a table with columns for 'Commit', 'Build #', and 'Build #'. The table lists three builds: '::UI XML Backup::', '::TFB Last(6-09-2018) 1.0.0.438.5ae7a3d76b3f289e04bdd39fe2...', and '::Sonar Fixes and Shipping Agent Invoice Search Screen::'.

2. select a source as **Azure Repos Git**, choose relevant **Team project**, **Repository** & **Default branch for manual and scheduled builds** and press **Continue**

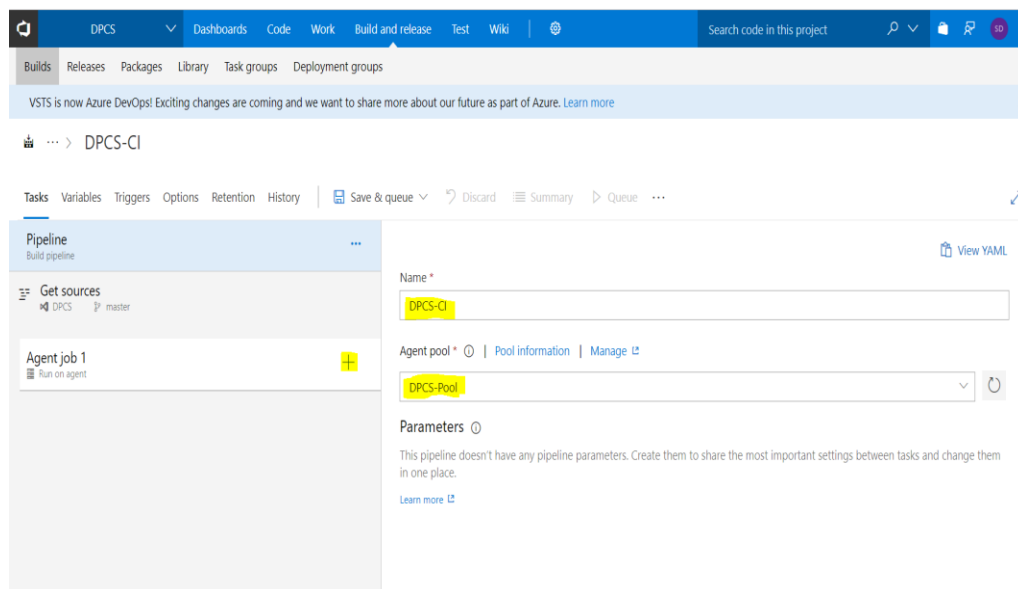


The screenshot shows the 'Select a source' dialog box in VSTS. The 'Azure Repos Git' option is selected. The 'Team project' dropdown is set to 'DPCS', the 'Repository' dropdown is set to 'DPCS', and the 'Default branch for manual and scheduled builds' dropdown is set to 'master'. The 'Continue' button is highlighted.

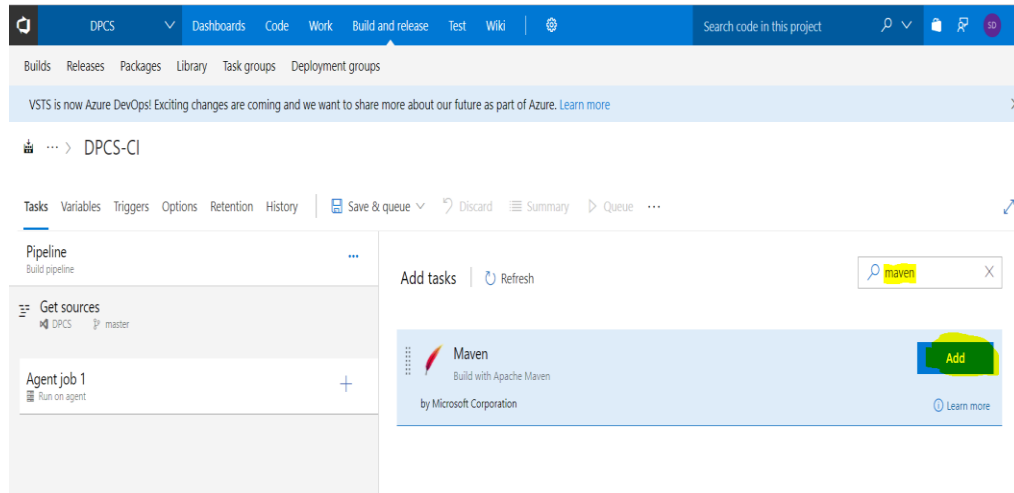
3. click on **Empty job**



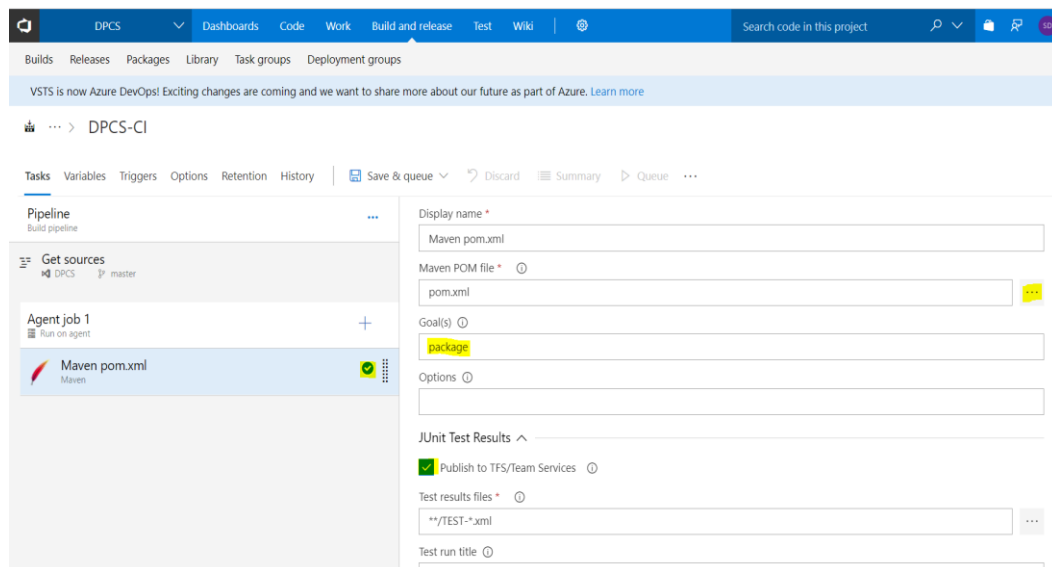
4. enter the **Name** for the job, type the **Agent pool** that you have created in the project and then click on **+ icon** to add task to the agent job.



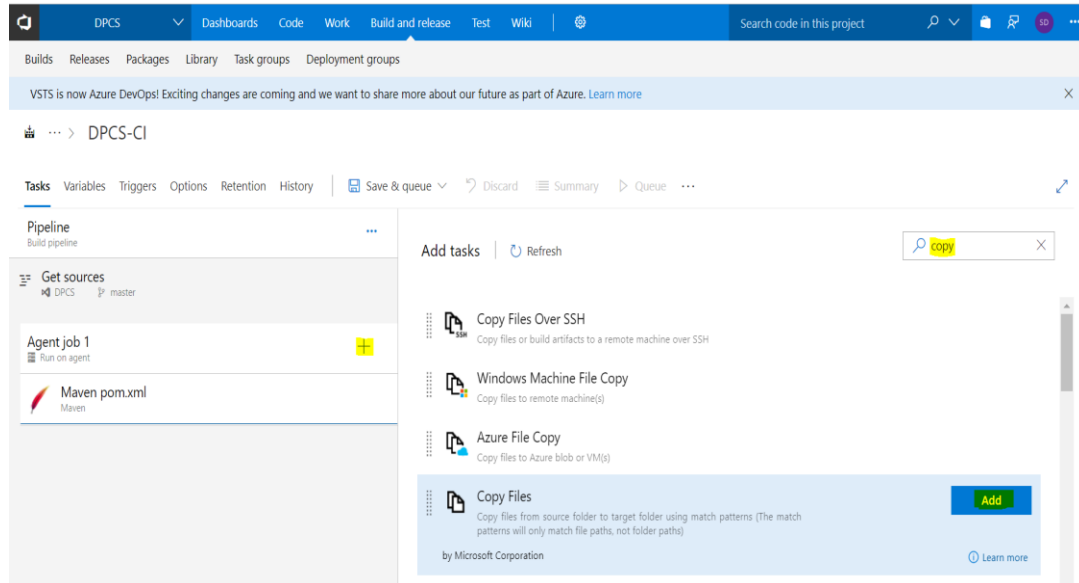
5. search for maven and Add to the task list



6. once the maven task is added select maven task and enter the below details
- choose the pom.xml from the source code by click on the 3 dots highlighted in the below image
 - enter your project Goal(s), by default it will be **package**. If needed you can change to **clean install** (based on in which way you need to build the source code)
 - If needed to run the JUnit Test Results, select the **Publish to TFS/Team Services** option or else unselect for not to run the JUnit Test Results.

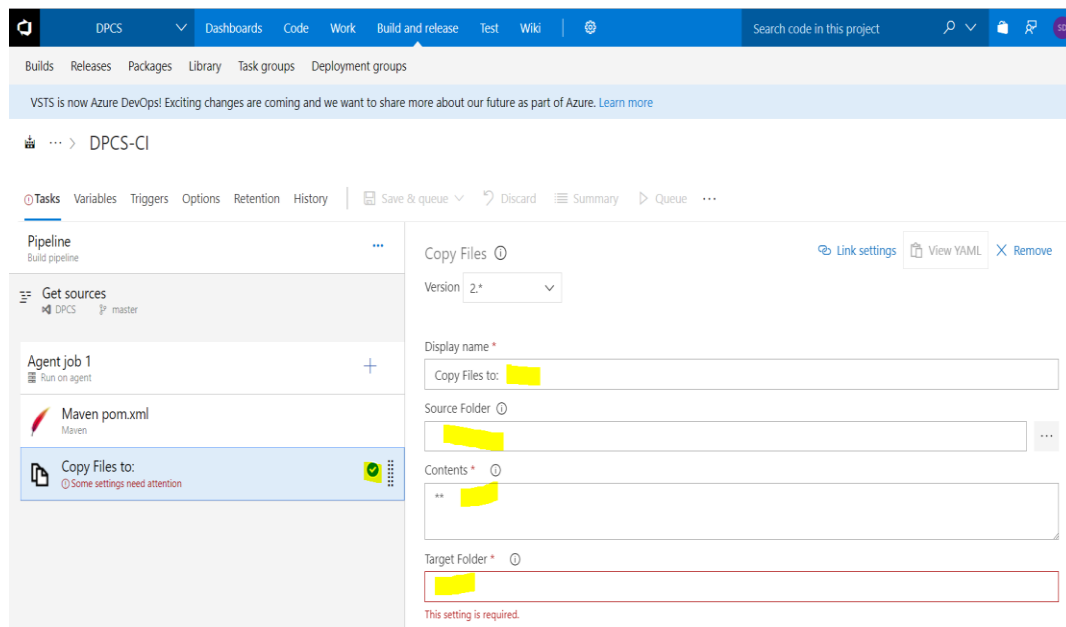


7. add a new task by click on + icon, type copy in the search box and select Copy Files option and add it to the task list



The screenshot shows the VSTS interface for a pipeline named 'DPCS-CI'. The 'Add tasks' panel is open, displaying a search bar with 'copy' entered. The 'Copy Files' task is selected, showing its description: 'Copy files from source folder to target folder using match patterns (The match patterns will only match file paths, not folder paths)'. The task is attributed to 'Microsoft Corporation' and has an 'Add' button.

8. select Copy Files task and enter the below details
 Display name: **Copy Files to: \$(build.artifactstagingdirectory)**
 Source Folder: **\$(system.defaultworkingdirectory)**
 Contents: ****/*.war**
 Target Folder: **\$(build.artifactstagingdirectory)**



The screenshot shows the VSTS interface for the 'Copy Files' task configuration. The task is named 'Copy Files to:' and has a version of '2.*'. The configuration fields are as follows:

- Display name ***: Copy Files to: [redacted]
- Source Folder**: [redacted]
- Contents ***: ** [redacted]
- Target Folder ***: [redacted]

A red border and the text 'This setting is required.' are visible under the Target Folder field, indicating it is mandatory.

Tasks Variables Triggers Options Retention History | Save & queue Discard Summary Queue ...

Pipeline Build pipeline

Get sources DPCS master

Agent job 1 Run on agent

Maven DEAIIP-Interoperability/pom.xml Maven

Copy Files to: \$(build.artifactstagingdirecto... Copy Files

Publish Artifact drop Publish Build Artifacts

Version 2.*

Display name * Copy Files to: \$(build.artifactstagingdirectory)

Source Folder \$(system.defaultworkingdirectory)

Contents * **/*.war

Target Folder * \$(build.artifactstagingdirectory)

9. click on + icon to add another task. Type **publish** in the search box and select **Publish Build Artifacts**

DPCS Dashboards Code Work Build and release Test Wiki | Search code in this project

Builds Releases Packages Library Task groups Deployment groups

VSTS is now Azure DevOps! Exciting changes are coming and we want to share more about our future as part of Azure. [Learn more](#)

DPCS-CI

Tasks Variables Triggers Options Retention History | Save & queue Discard Summary Queue ...

Pipeline Build pipeline

Get sources DPCS master

Agent job 1 Run on agent

Maven pom.xml Maven

Copy Files to: \$(build.artifactstagingdirecto... Copy Files

Add tasks Refresh

PyPI Publisher Create and upload an sdist or wheel to a PyPI-compatible index using Twine.

Publish Build Drop Publish Build Drop

Publish Test Results Publish Test Results to VSTS/TFS

Index Sources & Publish Symbols Index your source code and publish symbols to a file share or Visual Studio Team Services Symbol Server

Publish Build Artifacts Publish build artifacts to Visual Studio Team Services/TFS or a file share [Add](#)

by Microsoft Corporation [Learn more](#)

10. click on Publish Build Artifacts and leave the default values

DPCS Dashboards Code Work Build and release Test Wiki | Search code in this project

Builds Releases Packages Library Task groups Deployment groups

VSTS is now Azure DevOps! Exciting changes are coming and we want to share more about our future as part of Azure. [Learn more](#)

DPCS-CI

Tasks Variables Triggers Options Retention History | Save & queue Discard Summary Queue ...

Pipeline Build pipeline

Get sources DPCS master

Agent job 1 Run on agent

Maven pom.xml Maven

Copy Files to: \$(build.artifactstagingdirecto... Copy Files

Publish Artifact: drop Publish Build Artifacts

Version 1.*

Display name * Publish Artifact: drop

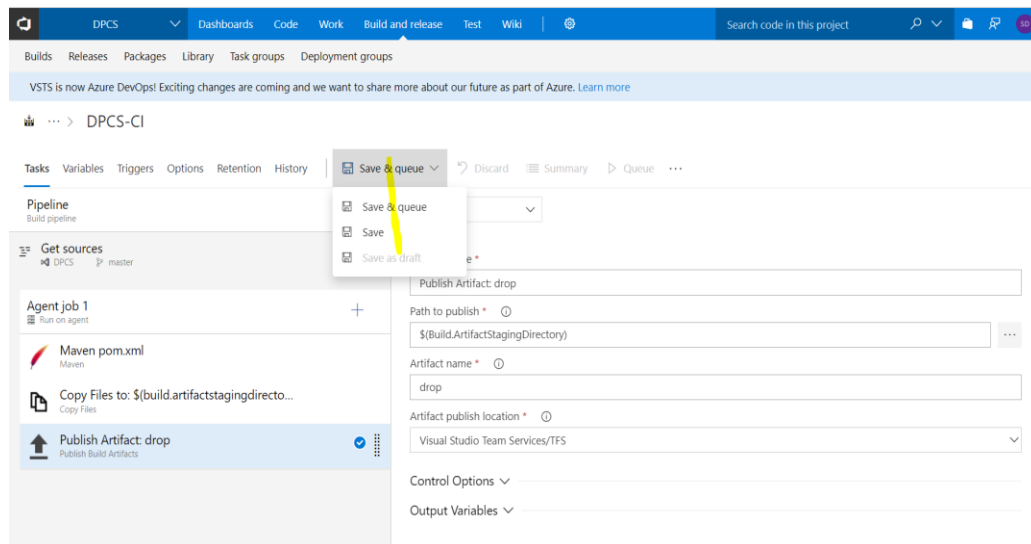
Path to publish * \$(Build.ArtifactStagingDirectory)

Artifact name * drop

Artifact publish location * Visual Studio Team Services/TFS

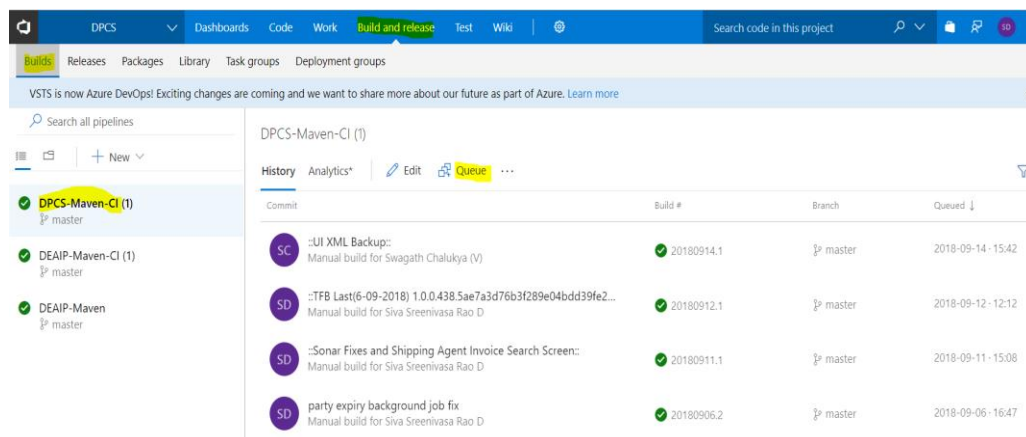
Control Options

11. click on Save or Save & Queue option to trigger the manual build.



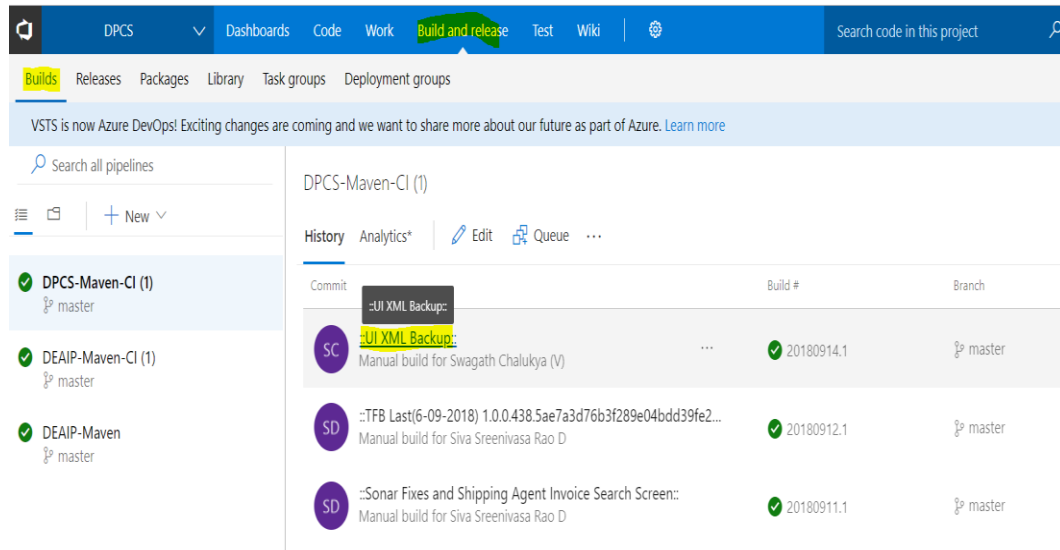
5. STEPS TO TRIGGER JOBS MANUALLY IN VSTS

1. click on **Build and release** → **Builds** → select your job → **Queue**



6. STEPS TO CHECK JOBS EXECUTION AND LOGS IN VSTS

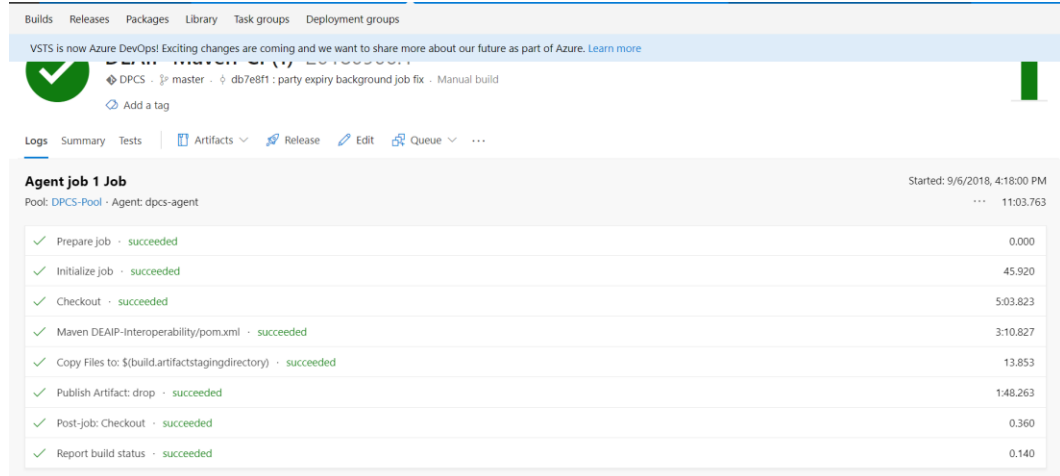
1. click on **Build and release** → **Builds** → select your job → click on the **completed/running job**



The screenshot shows the VSTS interface with the 'Build and release' tab selected. Under 'Builds', a list of build jobs is displayed for the 'DPCS-Maven-CI (1)' master branch. The jobs include:

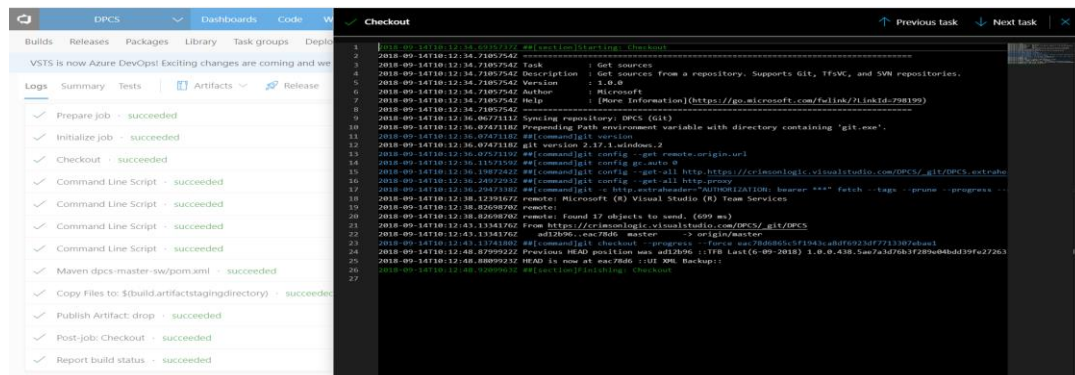
- Build # 20180914.1: Manual build for Swagath Chalukya (V)
- Build # 20180912.1: Manual build for Siva Sreenivasa Rao D
- Build # 20180911.1: Manual build for Siva Sreenivasa Rao D

2. click on **Logs** tab and click on which sequence job logs intended to see.



The screenshot shows the 'Logs' tab for 'Agent job 1 Job'. A table lists the build steps and their durations:

Step	Duration
Prepare job - succeeded	0.000
Initialize job - succeeded	45.920
Checkout - succeeded	503.823
Maven DEAIIP-Interoperability/pom.xml - succeeded	3:10.827
Copy Files to: \$(build.artifactstagingdirectory) - succeeded	13.853
Publish Artifact: drop - succeeded	1:48.263
Post-job: Checkout - succeeded	0.360
Report build status - succeeded	0.140



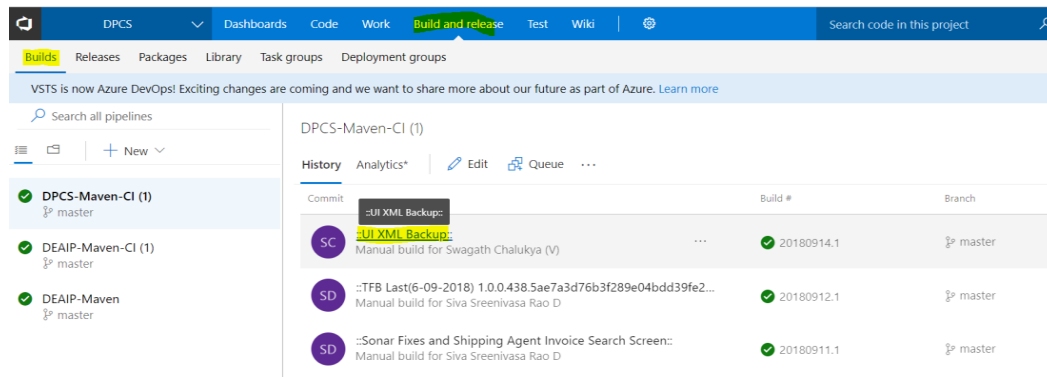
The screenshot shows the 'Checkout' step logs for 'Agent job 1 Job'. The logs indicate that the checkout was successful and the build was triggered by a manual build for Siva Sreenivasa Rao D.

```

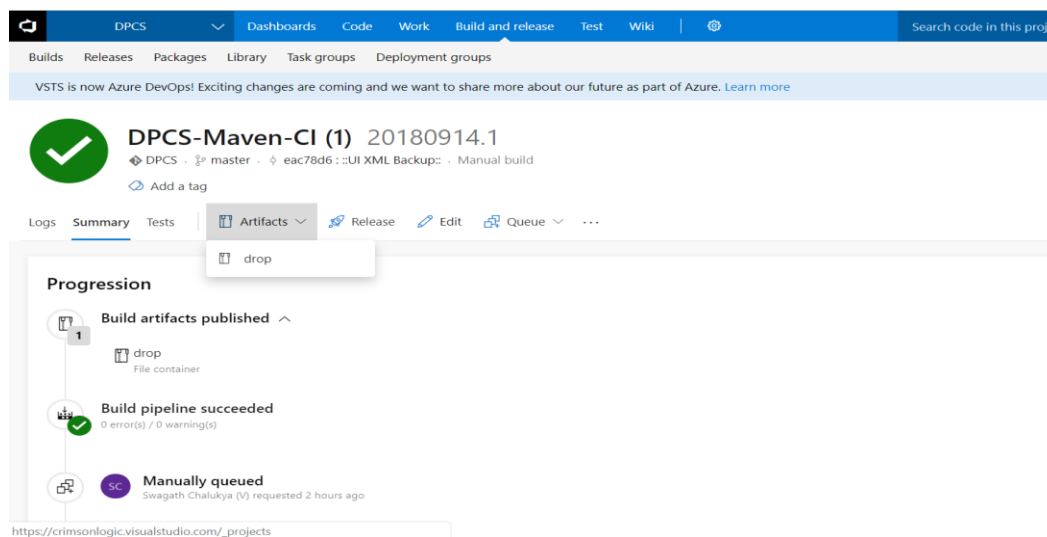
1 [Info] 2018-09-14T10:12:14.7195754Z #[command] Starting checkout
2 2018-09-14T10:12:14.7195754Z Task : Get sources
3 2018-09-14T10:12:14.7195754Z Description : Get sources from a repository. Supports Git, TFSVC, and SVN repositories.
4 2018-09-14T10:12:14.7195754Z Version : 3.0.0
5 2018-09-14T10:12:14.7195754Z Author : Microsoft
6 2018-09-14T10:12:14.7195754Z Help : [More Information](https://go.microsoft.com/fwlink/?linkid=798195)
7
8 2018-09-14T10:12:14.7195754Z =====
9 2018-09-14T10:12:14.7195754Z Spring repository: DPCS (Git)
10 2018-09-14T10:12:14.7195754Z Prepending Path environment variable with directory containing 'git.exe'.
11 2018-09-14T10:12:14.7195754Z #[command] git checkout
12 2018-09-14T10:12:14.7195754Z git version 2.17.1.windows.2
13 2018-09-14T10:12:14.7195754Z #[command] git config --get remote.origin.url
14 2018-09-14T10:12:14.7195754Z #[command] git config --get all --file=https://crimsonlogic.visualstudio.com/DPCS/_git/DPCS-external
15 2018-09-14T10:12:14.7195754Z #[command] git config --get all --file=https://crimsonlogic.visualstudio.com/DPCS/_git/DPCS-external
16 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
17 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
18 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
19 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
20 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
21 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
22 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
23 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
24 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
25 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
26 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
27 2018-09-14T10:12:14.7195754Z #[command] git checkout --progress --force
  
```

7. STEPS TO DOWNLOAD THE BUILD ARTIFACTS FROM VSTS

1. click on **Build and release** → **Builds** → select your job → click on the successfully completed job



2. Click on **Artifacts** tab and choose **drop** option



3. In the **Artifacts explorer** popup, navigate down to the target folder and check for the artifacts. Artifacts can be downloaded or can copy download URL.

